

# ELTA: An Enhancer against Long-Tail for Aesthetics-oriented Models

ICML 2024 – Summary

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# Problem

- **Image Aesthetics Assessment (IAA)** datasets are **long-tailed**
  - Most images → medium scores
  - Few images → very low / very high scores
- Consequences:
  - Models biased to majority
  - Poor prediction for minority samples
  - Fairness and generalization compromised

# Challenges in IAA

## 1. No clear class boundaries

- Scores are continuous, subjective

## 2. Feature-label mismatch

- Same image can have different opinions
- Raw-pixel augmentations may break aesthetics

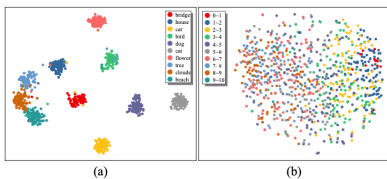


Figure 2: T-SNE visualization of feature distributions extracted by a classification network and an IAA network respectively. The entire score range in IAA is divided into ten segments, with feature points in each segment marked in different colors. The classification features (a) clearly distinguish between different categories, while the IAA features (b) exhibit noticeable confusion. The details are further explained in [supplement material A.3](#).

## 3. Smooth output distributions

- Hard to select reliable pseudo-labels

# ELTA Overview

Three plug-and-play modules:

1. **TFA** – Tail Features Augmentation
2. **FLSA** – Feature-Label Similarity Alignment
3. **APDS** – Adaptive Probability Distribution Sharpening

# Problem Setup

- Two tasks in **Image Aesthetics Assessment (IAA)**:

## 1) Distribution Prediction

- Training set:  $D = \{(x_i, y_i)\}_{i=1}^N$
- $x_i$ : input image
- $y_i = [y_i^1, y_i^2, \dots, y_i^C]$ : empirical PMF of human ratings
- Goal: predict  $\hat{y}_i$  close to  $y_i$
- Mean Opinion Score (MOS):
  - Ground-truth:  $s_i = \sum_{j=1}^C j \cdot y_i^j$
  - Predicted:  $\hat{s}_i = \sum_{j=1}^C j \cdot \hat{y}_i^j$

## 2) Score Regression

- Each image has scalar score  $y_i$
- Model may still output distribution, then compute predicted score  $\hat{s}_i$
- Loss minimized between  $\hat{s}_i$  and  $s_i$

# Long-Tail Challenge

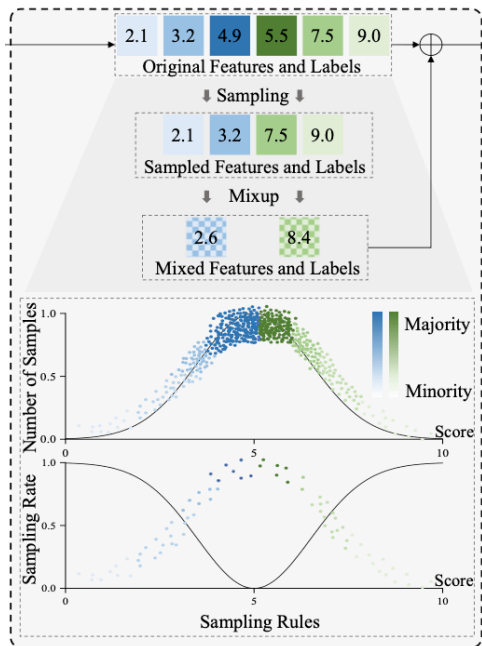
- IAA datasets → **long-tailed distributions**
  - Most images cluster around the mean  $\bar{s}$
  - Minority = very low or very high scores
- Consequence:
  - Model tends to predict medium scores
  - Error grows for minority:  $|\hat{s}_i - s_i| \propto |s_i - \bar{s}|$

Core focus: **reduce amplified errors on minority samples**

# Tail Features Augmentation (TFA)

- Why feature-level?

- Pixel-level augmentation (crop, flip, rotate) may ruin aesthetics
- Feature-level mixup preserves aesthetics while generating diverse samples



(a) TFA (optional)

# Mixup Formulation

Given two samples  $(x_i, y_i), (x_j, y_j)$ :

- Extract features at layer  $k$ :

$$z_i^k = f^k(x_i), z_j^k = f^k(x_j)$$

- Mixup:

$$\tilde{z}^k = \lambda z_i^k + (1 - \lambda) z_j^k$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

- When  $k = 0$ : equivalent to pixel-level mixup



# Sampling Strategy

- **Problem:** Random sampling → dominated by majority → little benefit to minority
- **Solution:** Two-stage sampling

Stage 1: Choose first sample  $z_i$

$$P(i) = \frac{\exp(|\bar{s} - s_i|/\tau_1)}{\sum_{k=1}^B \exp(|\bar{s} - s_k|/\tau_1)}$$

- Farther from mean → higher probability
- Bias toward minority

# Sampling Strategy (cont.)

Stage 2: Choose pair  $z_j$

$$P(j|i) = \frac{\exp(\tau_2/|s_i - s_j|)}{\sum_{k \neq i} \exp(\tau_2/|s_i - s_k|)}$$

- Smaller score difference  $\rightarrow$  higher probability
- Ensures aesthetics consistency in pairs

# Improved Mixup Coefficient

Traditional mixup:  $\lambda$  from Beta distribution

ELTA redefines:

$$\lambda = \frac{P(i)}{P(i)+P(j)} = \frac{\exp(|\bar{s}-s_i|/\tau_1)}{\exp(|\bar{s}-s_i|/\tau_1)+\exp(|\bar{s}-s_j|/\tau_1)}$$

Sample closer to minority gets **higher weight**

- Mixed feature dominated by minority
- Reduces accidental dominance of majority

# Feature-Label Similarity Alignment (FLSA)

- **Motivation:**

- IAA features often confused → mismatch with labels
- Principle: “Closer in label space → closer in feature space”



# FLSA Formulation

## 1. Feature similarity matrix

$$M_{i,j}^f = \cos(z_i, z_j) = \frac{z_i \cdot z_j}{|z_i||z_j|}$$

## 2. Label similarity matrix

$$M_{i,j}^l = 1 - \frac{|s_i - s_j|}{\max_k s_k - \min_k s_k}$$

- $M_{i,j}^l = 1$ : identical scores
- $M_{i,j}^l = 0$ : largest score gap

# FLSA Loss

- Alignment loss:

$$\mathcal{L}_{align} = \text{MSE}(M^f, M^l)$$

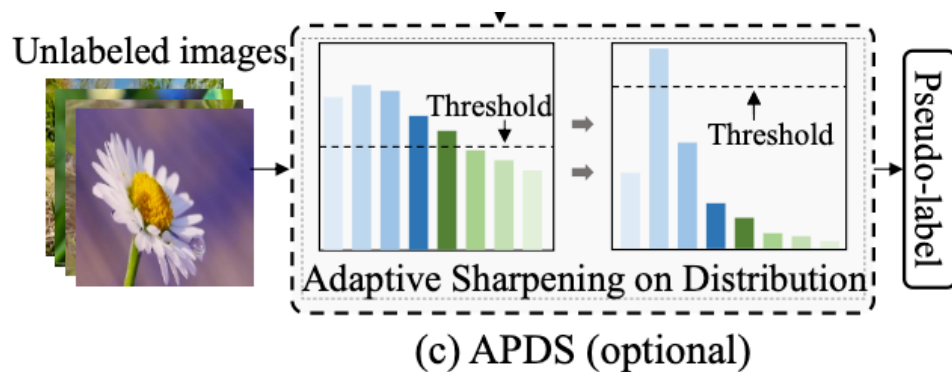
- Added as regularization:

$$\mathcal{L} = \mathcal{L}_{sup} + \alpha \cdot \mathcal{L}_{align}$$

Forces feature space to follow label similarity → better representation

# Adaptive Probability Distribution Sharpening (APDS)

- **Problem:** IAA predictions too smooth
  - Hard to set threshold for pseudo-label selection
- **Goal:** Sharpen minority predictions → more likely to pass threshold



# Temperature Scaling

- Given logits  $z_i$ :

$$\hat{y}_i = \text{Softmax}(z_i/\tau)$$

- $\tau > 1$ : smoother
- $\tau < 1$ : sharper



# Adaptive Temperature

Define sample-specific temperature:

$$\tau_i(\beta) = e^{-\beta|\hat{s}_i - \bar{s}|}$$

- Minority (far from mean)  $\rightarrow \tau_i \rightarrow 0 \rightarrow$  sharper
- Majority (close to mean)  $\rightarrow$  smoother

# Pseudo-label Selection

- Keep samples if:

$$\max(\hat{y}_i(\beta)) \geq t$$

- **Grid search** finds best  $(\beta^*, t^*)$  by minimizing MAE:

$$(\beta^*, t^*) = \arg \min_{\beta, t} \sum |\hat{s}_i(\beta, t) - s_i|$$

Ensures high-quality pseudo-labels

# Overall Architecture

- **Backbone:** Swin Transformer V2
- **Modules:** TFA, FLSA, APDS (independent, plug-and-play)

Flow:

1. Input → Swin Transformer features
2. TFA augments minority features
3. FLSA aligns features with labels
4. APDS sharpens predictions → pseudo-labels
5. Combine labeled + pseudo-labeled data for training

# Overall Architecture

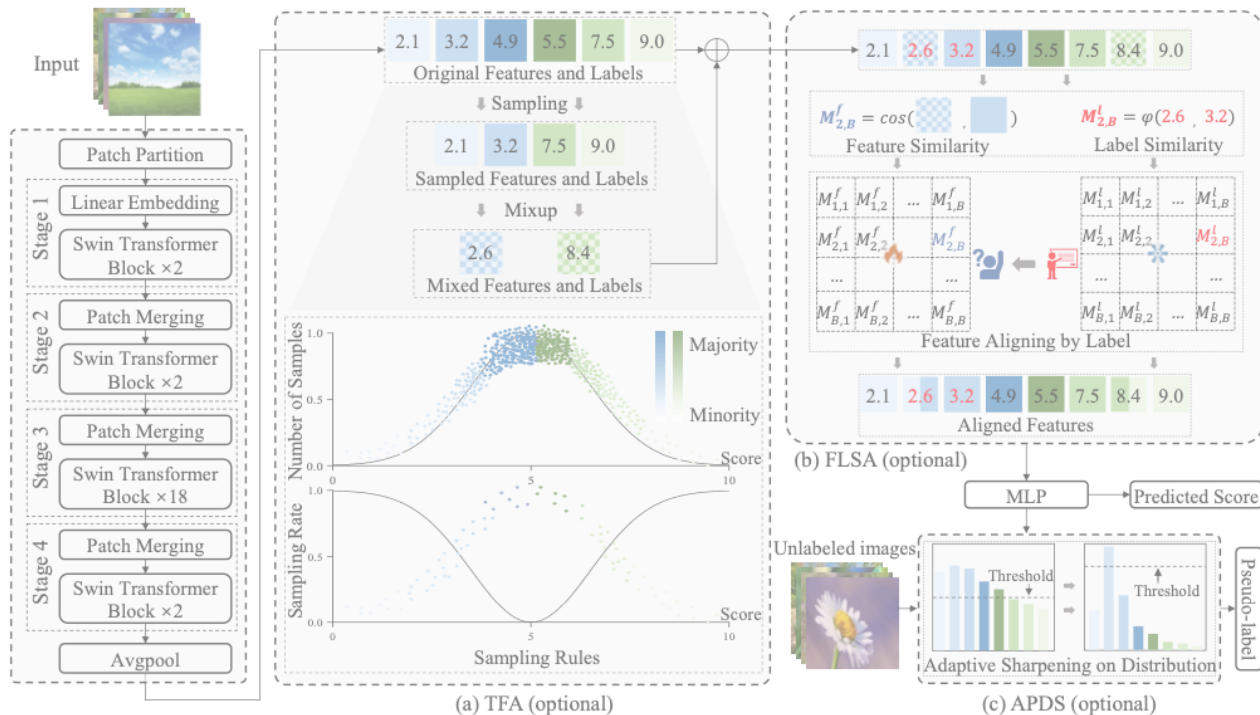


Figure 3: Architecture of the proposed ELTA model. ELTA consists of three **optional** components: (a) TFA performs mixup between instances sampled by our dedicated strategy to enhance minority features; (b) FLSA aligns features and labels based on their similarity consistency; (c) APDS reshapes the predictive distributions to work together with high thresholding for high-quality pseudo-label selection.

# Training Loss

## 1. Labeled Loss ( $L_s$ )

$$L_s = \frac{1}{B_l} \sum H(y_i^l, f(x_i^l))$$

- EMD for distribution labels (e.g. AVA)
- MSE for scalar labels

# Training Loss (cont.)

## 2. Unlabeled Loss ( $L_u$ )

$$L_u = \frac{1}{B_u} \sum \mathbf{1}[\max(\hat{y}_i^u(\beta^*)) \geq t^*] \cdot H(\hat{y}_i^u, f(x_i^u))$$

- Only high-confidence pseudo-labels are used

## 3. Final Loss

$$L = L_s + L_u$$

# Summary

- **TFA:** Feature-level mixup → generate strong minority features
- **FLSA:** Align feature similarity with label similarity → reduce mismatch
- **APDS:** Adaptive sharpening → reliable pseudo-labels
- **Training:** Labeled + pseudo-labeled loss → enhance performance on long-tail

# Module Comparison

| Module      | Problem                    | Method                                       | Effect                     |
|-------------|----------------------------|----------------------------------------------|----------------------------|
| <b>TFA</b>  | Minority under-represented | Feature-level mixup, biased sampling         | Enhance minority features  |
| <b>FLSA</b> | Feature-label mismatch     | Align cosine similarity (features vs labels) | Consistent representation  |
| <b>APDS</b> | Smooth predictions         | Adaptive temperature scaling                 | High-quality pseudo-labels |



# Experimental Results

**Datasets:** AVA, AADB, TAD66K, PARA

**Metrics:** PLCC, SRCC, MAE (Low/Medium/High)

Table 1: Comparing our ELTA with 7 IAA methods on the four datasets. The 'Base.' represents a Swin Transformer baseline model removing all the modules proposed. The best result for each metric is bolded in **red**.

| Dataset | Metric                   | CNN-based models |       |       |              | Transformer-based models |       |              |       |              |
|---------|--------------------------|------------------|-------|-------|--------------|--------------------------|-------|--------------|-------|--------------|
|         |                          | NIMA             | HGCN  | BIAA  | TANet        | MaxViT                   | MUSIQ | EAT          | Base. | Ours         |
| AVA     | $\mathcal{P} \uparrow$   | 0.636            | 0.687 | 0.668 | 0.765        | 0.745                    | 0.738 | 0.770        | 0.743 | <b>0.777</b> |
|         | $\mathcal{S} \uparrow$   | 0.612            | 0.665 | 0.651 | 0.758        | 0.708                    | 0.726 | 0.759        | 0.735 | <b>0.764</b> |
|         | $\mathcal{L} \downarrow$ | 0.655            | 0.675 | 0.653 | 0.630        | 0.600                    | 0.647 | 0.490        | 0.616 | <b>0.438</b> |
|         | $\mathcal{M} \downarrow$ | 0.322            | 0.321 | 0.382 | <b>0.237</b> | 0.317                    | 0.305 | 0.313        | 0.295 | 0.302        |
|         | $\mathcal{H} \downarrow$ | 0.648            | 0.660 | 0.568 | 0.729        | 0.531                    | 0.628 | 0.433        | 0.513 | <b>0.426</b> |
|         |                          |                  |       |       |              |                          |       |              |       |              |
| AADB    | $\mathcal{P} \uparrow$   | 0.711            | 0.734 | 0.733 | 0.742        | 0.748                    | 0.761 | 0.767        | 0.740 | <b>0.772</b> |
|         | $\mathcal{S} \uparrow$   | 0.700            | 0.716 | 0.710 | 0.749        | 0.742                    | 0.751 | 0.759        | 0.732 | <b>0.760</b> |
|         | $\mathcal{L} \downarrow$ | 1.450            | 1.453 | 1.508 | 1.394        | 1.592                    | 1.447 | 1.375        | 1.526 | <b>1.289</b> |
|         | $\mathcal{M} \downarrow$ | 0.874            | 0.989 | 0.897 | 0.846        | <b>0.782</b>             | 0.880 | 0.828        | 0.896 | 0.905        |
|         | $\mathcal{H} \downarrow$ | 1.527            | 1.299 | 1.423 | 1.355        | 1.461                    | 1.159 | 1.260        | 1.402 | <b>1.141</b> |
|         |                          |                  |       |       |              |                          |       |              |       |              |
| TAD66K  | $\mathcal{P} \uparrow$   | 0.405            | 0.493 | 0.431 | 0.531        | 0.513                    | 0.517 | <b>0.546</b> | 0.507 | 0.539        |
|         | $\mathcal{S} \uparrow$   | 0.390            | 0.486 | 0.417 | 0.513        | 0.484                    | 0.489 | <b>0.517</b> | 0.478 | 0.496        |
|         | $\mathcal{L} \downarrow$ | 1.851            | 1.808 | 1.734 | 1.598        | 1.570                    | 1.627 | 1.591        | 1.621 | <b>1.457</b> |
|         | $\mathcal{M} \downarrow$ | 0.812            | 0.780 | 0.876 | <b>0.682</b> | 0.746                    | 0.728 | 0.782        | 0.793 | 0.812        |
|         | $\mathcal{H} \downarrow$ | 1.690            | 1.370 | 1.669 | 1.651        | 1.402                    | 1.460 | 1.175        | 1.354 | <b>1.162</b> |
|         |                          |                  |       |       |              |                          |       |              |       |              |
| PARA    | $\mathcal{P} \uparrow$   | 0.862            | 0.881 | 0.886 | 0.899        | 0.936                    | 0.918 | 0.940        | 0.925 | <b>0.943</b> |
|         | $\mathcal{S} \uparrow$   | 0.877            | 0.865 | 0.858 | 0.887        | 0.902                    | 0.899 | 0.909        | 0.897 | <b>0.912</b> |
|         | $\mathcal{L} \downarrow$ | 0.616            | 0.573 | 0.469 | 0.551        | 0.383                    | 0.572 | 0.336        | 0.402 | <b>0.327</b> |
|         | $\mathcal{M} \downarrow$ | 0.344            | 0.290 | 0.328 | 0.299        | 0.282                    | 0.315 | 0.276        | 0.314 | <b>0.251</b> |
|         | $\mathcal{H} \downarrow$ | 0.486            | 0.502 | 0.503 | 0.429        | 0.276                    | 0.424 | <b>0.256</b> | 0.379 | 0.290        |
|         |                          |                  |       |       |              |                          |       |              |       |              |

ELTA improves correlation and reduces minority errors across all datasets.

# Contributions

- First work addressing **long-tail problem in IAA**
- Proposed **ELTA** with 3 modules:
  - Enhance minority features
  - Align features & labels
  - Sharpen distributions for pseudo-labels
- Plug-and-play, improves existing methods
- Achieves SOTA on 4 benchmark datasets

Thank you for listening!